

2025, Fall

1. a). Show that $\phi: P_n(\mathbb{R}) \rightarrow \mathbb{R}: p(t) \mapsto p'(0)$ is linear functional.

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b). Find a dual basis corresponding to $\{1, t, \dots, t^n\}$ of $P_n(\mathbb{R})$.

Proof. Let $f(x) = \sum_{i=0}^n a_i x^i$ and $g(x) = \sum_{i=0}^n b_i x^i$ and $C \in \mathbb{R}$ be given.

Then, $\phi(Cf + g) = (Cf + g)'|_{t=0} = C \cdot f'|_{t=0} + g'|_{t=0} = C \cdot \phi(f) + \phi(g)$

by the linearity of differentiation. This shows the a),

To show b), let

$$\mathcal{B}^* := \left\{ \frac{1}{k!} \cdot \phi_k: P_n(\mathbb{R}) \rightarrow \mathbb{R}: p(t) \mapsto p^{(k)}(0) \mid k=0, 1, \dots, n \right\}$$

Then, for each $k=0, 1, \dots, n$,

$$\frac{1}{k!} \cdot \phi_k(t^k) = \frac{k!}{k!} \cdot 1 = 1, \text{ and } \frac{1}{k!} \cdot \phi_k(t^r) = 0 \text{ if } r \neq k.$$

Similarly a), the elements in $\mathcal{B}^*$ are all linear functional, so the $\mathcal{B}^*$ is dual basis.

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2. Let $A \in M_{\text{max}}(1\mathbb{R})$.

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a). For $n \in \mathbb{N}$, $\text{rank } A^n \leq \text{rank } A$.

b). Let $L_A: \mathbb{R}^m \rightarrow \mathbb{R}^m, x \mapsto Ax$.

Show that for $n \geq 2$,

$\text{rank } A^n = \text{rank } A$ implies $\ker L_A \cap \text{Im } L_A = \{0\}$.

Sol) a). Given $x \in \text{Im } A^n$, $x = A^n y$ for some $y \in \mathbb{R}^m$.

So, $x = A^n y = A^{n-1} A y \in \text{Im } A$, for $n \geq 2$. Thus $\text{Im } A^n \subseteq \text{Im } A$, it follows $\text{rank } A^n \leq \text{rank } A$. If $n=1$, clear.

b). Given $x \in \ker A$, $A^n x = A^{n-1} A x = A^{n-1} 0 = 0$, so $x \in \ker A^n$, so $\ker A \subseteq \ker A^n$. Since $\text{rank } A^n = \text{rank } A$, the dimension theorem gives $\text{nullity } A^n = \text{nullity } A$.

Therefore, since $\mathbb{R}^m$ is finite dimensional, so is $\ker A^n$, thus $\ker A = \ker A^n$.

Now, let $x \in \ker L_A \cap \text{Im } L_A$ be given. Then, $Ax = 0$ and $x = Ay$ for some $y \in \mathbb{R}^m$.

Then, $A^{n-1} x = A^n y = 0$, because $Ax = 0$, so $A^{n-2} Ax = 0$, and this implies $y \in \ker A^n$.

Since $\ker A^n = \ker A$, $Ay = 0$, so $x = Ay = 0$, so proved. $\square$

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3. Find all $a, b, c \in \mathbb{R}$ such that

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$$A = \begin{bmatrix} 2 & 0 & 0 \\ a & 2 & 0 \\ b & c & -1 \end{bmatrix} \text{ is diagonalizable.}$$

Sol. Since the characteristic polynomial of A is regardless of a, b, c ,

$$\chi_A(t) = (t-2)^2(t+1).$$

Since a Matrix is diagonalizable if and only if the minimal polynomial splits and separable, and the minimal polynomial divides the characteristic polynomial, the A is diagonalizable if and only if

$$(A-2I)(A+I) = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & -3 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 & 0 \\ a & 3 & 0 \\ b & c & 0 \end{bmatrix} = 0.$$

$$\text{Calculating gives } = \begin{bmatrix} 0 & 0 & 0 \\ 3a & 0 & 0 \\ ac & 0 & 0 \end{bmatrix}$$

therefore, $a=0$, $b \in \mathbb{R}$, $c \in \mathbb{R}$ are all possible (a, b, c) .

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4. Let $N \subseteq GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be the set of all pairs (A, B) such that $\det A = \pm \det B$.

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Show that N is normal and determine the factor group $(G \times G)/N$.

Sol). Let $(A, B) \in N$ and $(C, D) \in GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be given. Then

$$(C, D)(A, B)(C, D)^{-1} = (CAC^{-1}, DBD^{-1})$$

And, since the determinant preserves the multiplication of matrix,

$$\det CAC^{-1} = \det C \cdot \det C^{-1} \cdot \det A = \det A$$

$$\text{and } \det DBD^{-1} = \det D \cdot \det D^{-1} \cdot \det B = \det B$$

$$\text{so, since } (A, B) \in N, \det CAC^{-1} = \det A = \pm \det B = \pm \det DBD^{-1}$$

therefore $(C, D)(A, B)(C, D)^{-1} \in N$. And, to show N is subgroup,

first, $(I, I) \in N$ trivially, so $N \neq \emptyset$. Let $(A, B), (C, D) \in N$ be given. Then

$$(A, B) \cdot (C, D)^{-1} = (AC^{-1}, BD^{-1})$$

$$\text{satisfies } \det AC^{-1} = \det A \cdot \det C^{-1} = \pm \det B \cdot \det C^{-1} = \pm \det B \cdot \det D^{-1} = \pm \det BD^{-1}$$

therefore, by the subgroup criterion, N is subgroup, and normal. $\perp$ hom.

Define a map

$$\mathcal{Q}: GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) \rightarrow \mathbb{R}^{\times}; (A, B) \mapsto (\det A - \det B)^2$$

Then, $\ker \mathcal{Q} = N$ and $\mathcal{Q}$ is homomorphism, onto $(0, \infty)$, because

the det preserves multiplication, so is $\mathcal{Q}$, and onto is given by: fix $B = I$, and $A = \text{diag}(r, 1, \dots, 1)$ where $r > 0$ is arbitrary.

By the first isomorphism theorem,

$$GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) / N \cong ((0, \infty), \times) \xrightarrow{\perp} \text{hom.} \quad \textcircled{2}$$

Comment: $\mathcal{Q}$ 은 먼저 $\mathbb{R}^{\times}$ 라고 시작하고, hom 임을 보려면 앞 계산을 꼭 해야 함!

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5. Let G be an order 21 subgroup of $\mathbb{C}^\times$.

a) G is cyclic.

b). For any $g \in G$, $x^{12} = g^{15}$ has 3 distinct solutions in G .

Proof. a). Since $\mathbb{C}$ is abelian, so G is also abelian.

By the Cauchy's theorem, G has order 3 element x and order 7 element y .
Now, since 3 and 7 are relatively prime, $|xy| = 21$, so $G = \langle xy \rangle$.

b). By a), $G \cong \mathbb{Z}/21\mathbb{Z}$. So, we can write the given equation as

$$12x \equiv 15g \pmod{21} \quad (g \in \mathbb{Z}/21\mathbb{Z}).$$

Regardless of g , $\gcd(12, 21) = 3$ divides $15g$, so it has three distinct solutions.

Note: $ax \equiv b \pmod{n}$ has a solution iff $\gcd(a, n) \mid b$.

And in such a case, it has $\gcd(a, n)$ solutions.

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6. Let $\{a_n\}$ be a bounded sequence in $\mathbb{R}$. Define

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$$\{a_n\}^* = \{x \in \mathbb{R} \mid x \text{ is a subsequential limit of } \{a_n\}\}$$

a). Prove that $\{a_n\}^*$ is closed.

b). Show $\limsup_{n \rightarrow \infty} a_n = \sup \{a_n\}^*$

c). Is there a sequence $\{b_n\}$ such that $\{b_n\}^* = [0, 1]$?

Proof a). If $\{a_n\}^*$ is finite, then clearly closed in $\mathbb{R}$.

Assume that $\{a_n\}^*$ is infinite. Since $\{a_n\}$ is bounded, so the set of subsequential limits $\{a_n\}^*$ is bounded set. So it has a limit point.

Let α be a limit point of $\{a_n\}^*$. Then, given $\epsilon \in \mathbb{R}$, there exists a

subsequential limit p of $\{a_n\}$ such that $p \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$.

For $k=1$, choose such a $p_1 \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$, and since p is a subsequential limit, choose $a_{p_1} \in B_{\frac{1}{k}}(p)$, then $a_{p_1} \in B_{\frac{\epsilon}{2}}(\alpha)$.

For chosen $a_{p_1}, \dots, a_{p_{k-1}}$, choose $p_k \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$.

Then, similarly, there is $a_{p_k} \in B_{\frac{\epsilon}{2}}(\alpha)$ such that $p_k > p_{k-1}$.

because no such p_k exists, then it is contradiction to a_{p_k} is a subsequential limit. Therefore, the arbitrary limit point α is contained in $\{a_n\}^*$,

so it is closed.

b). Since a), $\{a_n\}^*$ is closed and bounded, so compact. Thus

$$\sup \{a_n\}^* = \max \{a_n\}^* = \alpha \text{ for some subsequential limit } \alpha.$$

So, given $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N \Rightarrow a_n < \alpha + \epsilon$ (since $\alpha = \max \{a_n\}^*$)

and there exists infinitely many terms in $\{a_n\}$ such that $\alpha - \epsilon < a_n$ (since $a_{p_k} \rightarrow \alpha$)

Thus $\limsup_{n \rightarrow \infty} a_n = \alpha$

c). Let b_n be defined as:

$$\frac{1}{1}, \frac{1}{2}, \frac{2}{2}, \frac{1}{3}, \frac{2}{3}, \frac{3}{3}, \dots$$

Then, given $b \in [0, 1]$, there exists a subsequence in $\{b_n\}$ converges to b , because: for any $\epsilon > 0$, there are infinitely many terms in b_n such that

$$|b_n - b| < \epsilon,$$

so we can construct the subsequence. (20)

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